

PUSALA PREETHAM

Software Engineering • Machine Learning • Full-Stack Development

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PROFILE

Computer Science undergraduate (GPA: 9.33/10) focused on Machine Learning and Full-Stack Engineering. Experienced in taking projects from data preparation and model development through APIs, databases, and user-facing applications. Built an AI-powered civic technology platform and a multi-horizon electricity forecasting pipeline using real-world smart-meter data. Strong foundation in Python, JavaScript, SQL, ML, backend systems, and DSA, with 367+ LeetCode problems solved.

EDUCATION

SRM University, Amaravati

2024 – 2028

B.Tech in Computer Science and Engineering | GPA: 9.33 / 10

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management, Operating Systems

RESEARCH EXPERIENCE

Research Intern — Electricity Consumption Forecasting (ML & DL)

Summer 2026

Time-series forecasting • Gradient boosting • Feature engineering • Weather-informed demand prediction

[GitHub](#)

- Designed and built an end-to-end multi-horizon forecasting pipeline on 3-minute smart-meter readings from 38 households (2019–2021), predicting electricity demand at 1-hour, 1-day, and 1-month horizons.
- Benchmarked XGBoost, LightGBM, and CatBoost across MAE, RMSE, SMAPE, and R^2 , achieving $R^2 = 0.93$ (daily) and 0.92 (monthly) with CatBoost, and 0.80 (hourly) with LightGBM.
- Engineered 20+ temporal and statistical features — rolling windows, exponential moving averages, trend and lag-aware signals, and calendar features — to capture consumption patterns.
- Enriched demand forecasts with live weather variables (temperature, humidity, precipitation, wind speed, surface pressure) via the Open-Meteo API.
- Ran feature-importance analysis to interpret model behavior and documented methodology and results using Python, Pandas, Matplotlib, and Scikit-learn.

PROJECTS

CivicSolve AI — AI-Powered Civic Problem-Solving Platform

[GitHub](#)

React, Node.js, Express.js, PostgreSQL, pgvector, Redis, Python, FastAPI, Ollama, Llama 3.2, Gemma 3

- Architected a multi-portal platform connecting citizens, government, students, universities, and industry to identify, verify, and collaboratively resolve real-world civic problems.
- Built AI-assisted processing pipelines using Llama 3.2 for structured problem classification and Gemma 3 for citizen-submitted image analysis, gated behind government verification before public access.
- Implemented semantic duplicate detection with Sentence Transformers and PostgreSQL pgvector, clustering similar citizen reports while preserving individual submissions and location data.
- Designed asynchronous AI workloads with Redis queues and PostgreSQL-backed workflows spanning reports, teams, faculty collaboration, and solution development.
- Built role-based REST APIs enabling secure portal-to-backend communication across citizen, student, university, industry, and government modules.

AI-Based Student Dropout Prediction System

[GitHub](#)

Python, Flask, Node.js, JavaScript, Scikit-learn, XGBoost

- Built an end-to-end ML application to identify at-risk students from academic and behavioral data, serving predictions through a Flask API and web interface.
- Preprocessed raw datasets, performed feature selection, and benchmarked Logistic Regression, Random Forest, and XGBoost to select the strongest predictive model.

TECHNICAL SKILLS

Languages: Python, JavaScript, C, SQL

ML / AI: Scikit-learn, XGBoost, LightGBM, CatBoost, Sentence Transformers, Ollama

Frontend: React, Tailwind CSS, HTML, CSS

Backend: Node.js, Express.js, Flask, FastAPI, REST APIs

Databases: PostgreSQL, pgvector, MySQL, MongoDB

Tools: Git, GitHub, Redis

ACHIEVEMENTS & ACTIVITIES

- Solved 367+ problems on LeetCode and 194 on CodeChef, reflecting consistent practice in algorithms, data structures, and competitive programming.
- Participated in Smart India Hackathon 2026, VibeCraft (HackSRM 7.0), VoyageHack 3.0, and the Sustainable Development Goals Hackathon, applying AI and full-stack skills under time-boxed, team-based conditions.